

How Local Averaging Moves Matérn Pairwise Range Targets: A Smoothness-Dependent Phase Law

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ABSTRACT

Spatial measurements often represent local averages but are fitted as point observations. This changes the population target of covariance estimation, not merely its finite-sample bias. We study a stationary Matérn Gaussian field in $\mathbb{R}^d$, observed through a known compact symmetric averaging kernel of bandwidth h . For a fixed nonzero-lag Gaussian two-point likelihood with fixed known smoothness ν , we characterize the exact variance and decay targets. The naive decay shift is of order $h^{2\nu}$ for $0 < \nu < 1$, $h^2 \log(1/h)$ at $\nu = 1$, and h^2 for $\nu > 1$. Explicit coefficients show that naive point-support fitting underestimates inverse range—and thus overestimates spatial range—for every $\nu > 0$ and any nondegenerate compact symmetric kernel when h is sufficiently small. The smooth-regime sign remains valid for anisotropic averaging. Deterministic two-dimensional quadrature verifies the three rate regimes. A 200-replicate finite-grid study then compares naive and support-aware full likelihoods, including boundary and irregular-design stress tests. The result isolates an inferential failure mode that geospatial simulation studies can reproduce and diagnose without an empirical dataset.

CCS CONCEPTS

• **Mathematics of computing** → **Probability and statistics**; *Probabilistic representations*.

KEYWORDS

Matérn covariance, change of support, Gaussian random field, misspecified likelihood, spatial simulation, range estimation

ACM Reference Format:

Sasha Cui. 2026. How Local Averaging Moves Matérn Pairwise Range Targets: A Smoothness-Dependent Phase Law. In *Proceedings of 9th ACM SIGSPATIAL International Workshop on Geospatial Simulation (GeoSim '26)*. ACM, New York, NY, USA, 5 pages.

1 INTRODUCTION

Values recorded on pixels, grid cells, or after kernel preprocessing are local aggregates of an underlying spatial field. Change-of-support models account for this by integrating the latent process over the measurement support [3, 6]. A common shortcut instead assigns each aggregate to a declared center and fits a point-support covariance. Relative to the latent parameters, the support is then misinterpreted. A single two-point covariance can still be matched exactly by different variance and decay parameters; a multi-lag likelihood is genuinely misspecified and selects a Kullback–Leibler (KL) projection [2].

This qualitative fact leaves three quantitative questions. How quickly does the pseudo-parameter move as support shrinks? Does the rate depend on Matérn smoothness? Is the direction of range

error stable across dimensions and averaging kernels? The answer is not an ordinary second-order Taylor expansion: the Matérn covariance changes regularity at the origin as smoothness crosses one.

Temporal aggregation of autoregressive and Ornstein–Uhlenbeck processes is classical [1, 4, 10]. Change-of-support reviews and area-to-point methods establish block covariance and prediction [7, 8]. Most closely, [5] builds a likelihood for block averages from an underlying point covariance and includes a Matérn example, but does not derive the parameter selected when those averages are instead treated as points. Generic misspecified-GP theory gives the KL framework but not this coefficient or sign [2]. We make no novelty claim for a one-dimensional exponential calculation or for $SKS^\top$. Our contribution is an all-smoothness, d -dimensional fixed-lag expansion of the naive Matérn pair-likelihood target:

- The leading inverse-range shift has order $h^{2\nu}$, $h^2 \log(1/h)$, or h^2 , depending on whether $\nu < 1$, $\nu = 1$, or $\nu > 1$.
- Explicit coefficients and a Bessel-function identity prove that the shift is negative for every $\nu > 0$, including under anisotropic symmetric averaging.
- A reproducible geospatial simulation protocol validates the continuous phase law and compares naive and support-aware full-grid likelihoods.

We use increasing-domain language only for repeated pair criteria. We do not claim separate fixed-domain consistency of Matérn variance and range, which would conflict with microergodicity results [11].

2 MODEL AND EXACT PAIR TARGET

Let $Y = \{Y(t) : t \in \mathbb{R}^d\}$ be a mean-zero stationary Gaussian field with

$$C(r) = v\mathcal{M}_\nu(\alpha\|r\|), \quad \mathcal{M}_\nu(x) = \frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_\nu(x), \quad (1)$$

where $v, \alpha, \nu > 0$ and $\mathcal{M}_\nu(0) = 1$. Thus α is inverse range. Let k be a symmetric probability density supported on $B(0, L)$, set $k_h(u) = h^{-d}k(u/h)$, and observe

$$Z_h(t) = \int k_h(u)Y(t-u)du. \quad (2)$$

For independent $U, V \sim k$, write $D = U - V$, $\Sigma_k = \mathbb{E}(UU^\top)$, $T_k = \text{tr}(\Sigma_k)$, and $m_q = \mathbb{E}\|D\|^q$. Then $m_2 = 2T_k$, and stationarity gives

$$C_h(r) = v\mathbb{E}[\mathcal{M}_\nu\{\alpha\|r + hD\|\}]. \quad (3)$$

Assume $T_k > 0$.

Fix $r = Re$, $R > 0$, and $\|e\| = 1$. A naive point-support pair model with known ν and unknown variance and inverse range uses

$$v^\dagger \begin{pmatrix} 1 & \mathcal{M}_\nu(\alpha^\dagger R) \\ \mathcal{M}_\nu(\alpha^\dagger R) & 1 \end{pmatrix}. \quad (4)$$

The smoothed two-by-two covariance lies in this family. Moreover, $0 < \rho_h(r) < 1$: positivity follows from $k \geq 0$ and $M_\nu > 0$, while strict inequality follows from the positive smoothed spectral density near the origin. Hence the exact Gaussian KL target is

$$v_h^\dagger = C_h(0), \quad M_\nu(\alpha_h^\dagger R) = \rho_h(r) := \frac{C_h(r)}{C_h(0)}. \quad (5)$$

It is unique because

$$M'_\nu(x) = -\frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_{\nu-1}(x) < 0. \quad (6)$$

On a fixed lattice and increasing rectangular Følner domains, a pair-composite estimator converges to this target under a compact parameter space, target uniqueness and interiority, correlations bounded away from one, and the usual continuity and integrable-envelope conditions for a uniform ergodic law. Dependence among overlapping pairs requires Godambe uncertainty; it does not change this population target.

3 SMOOTHNESS-DEPENDENT PHASE LAW

Put $z = \alpha R$, $a_k(e) = e^\top \Sigma_k e$, and

$$B_{\nu,k}(r) = \alpha^2 \left[a_k(e) M''_\nu(z) + \{T_k - a_k(e)\} \frac{M'_\nu(z)}{z} \right]. \quad (7)$$

PROPOSITION 1 (FIXED NONZERO LAG). *Uniformly for r in a compact annulus bounded away from zero, for $h \leq \|r\|/(4L)$, and for α in a compact subset of $(0, \infty)$,*

$$\frac{C_h(r)}{v} = M_\nu(\alpha \|r\|) + h^2 B_{\nu,k}(r) + O(h^4). \quad (8)$$

At zero lag, the Matérn origin expansion produces a phase transition.

THEOREM 1 (MATÉRN SUPPORT PHASE LAW). *Fix $\nu > 0$. Let*

$$c_\nu = \frac{\Gamma(1-\nu)}{v^{2\nu}\Gamma(\nu)} > 0, \quad 0 < \nu < 1.$$

Under the preceding assumptions, as $h \downarrow 0$,

$$\frac{C_h(0)}{v} = 1 - c_\nu m_{2\nu}(\alpha h)^{2\nu} + O(h^2), \quad 0 < \nu < 1, \quad (9)$$

$$= 1 - \frac{m_2}{2} (\alpha h)^2 \log\{1/(\alpha h)\} + O(h^2), \quad \nu = 1, \quad (10)$$

$$= 1 - \frac{m_2}{4(\nu-1)} (\alpha h)^2 + o(h^2), \quad \nu > 1. \quad (11)$$

The pseudo-decay in (5) satisfies

$$\alpha_h^\dagger - \alpha = \frac{M_\nu(z) c_\nu m_{2\nu} \alpha^{2\nu}}{R M'_\nu(z)} h^{2\nu} + O(h^{\min(4\nu, 2)}), \quad 0 < \nu < 1, \quad (12)$$

$$\alpha_h^\dagger - \alpha \sim \frac{m_2 M_1(z) \alpha^2}{2R M'_1(z)} h^2 \log(1/h), \quad \nu = 1, \quad (13)$$

$$\alpha_h^\dagger - \alpha = \frac{A_{\nu,k}(r)}{R M'_\nu(z)} h^2 + \mathcal{R}_\nu(h), \quad \nu > 1, \quad (14)$$

where

$$A_{\nu,k}(r) = \alpha^2 \{(2\nu-2)a_k(e) + T_k\} G_\nu(z) > 0, \quad (15)$$

$$G_\nu(z) = \frac{M_\nu(z)}{2\nu-2} + \frac{M'_\nu(z)}{z} \quad (16)$$

$$= \frac{2^{1-\nu}}{\Gamma(\nu)} \frac{z^\nu K_{\nu-2}(z)}{2\nu-2} > 0, \quad (17)$$

and $\mathcal{R}_\nu(h) = O(h^{2\nu})$ for $1 < \nu < 2$, $O(h^4 \log(1/h))$ for $\nu = 2$, and $O(h^4)$ for $\nu > 2$. For fixed ν , the remainders are uniform over compact positive decay sets and lag annuli bounded away from zero, but not as ν approaches one or two. Consequently, for every $\nu > 0$ and all sufficiently small $h > 0$,

$$v_h^\dagger < v, \quad \rho_h(r) > M_\nu(\alpha R), \quad \alpha_h^\dagger < \alpha.$$

Remark 1. The fitted lag R is fixed and bounded away from zero. If $R = ch$, the nonzero-lag expansion is no longer uniform and the relative displacement need not vanish.

Thus ignored support inflates inferred range. The sign is not an assumption: for $\nu > 1$, it follows from positivity of $K_{\nu-2}$ in the displayed identity. For an isotropic kernel, $A_{\nu,k}(r) = \sigma_k^2 \alpha^2 (2\nu + d - 2) G_\nu(z)$.

3.1 Proof sketch

Let $f_\alpha(x) = M_\nu(\alpha \|x\|)$. At a nonzero lag, fourth-order Taylor expansion is uniform. Symmetry and $\mathbb{E}(DD^\top) = 2\Sigma_k$ cancel odd terms, including the cubic term, and yield (8). At zero, the noninteger Bessel expansion is

$$M_\nu(x) = \sum_{j \geq 0} \frac{(x^2/4)^j}{j!(1-\nu)_j} + \frac{\Gamma(-\nu)x^{2\nu}}{2^{2\nu}\Gamma(\nu)} \sum_{j \geq 0} \frac{(x^2/4)^j}{j!(1+\nu)_j}. \quad (18)$$

Taking expectations at $x = \alpha h \|D\|$ proves (9) and (11). At $\nu = 1$,

$$M_1(x) = 1 + \frac{x^2}{2} \{\log(x/2) + \gamma_E - 1/2\} + O\{x^4(1 + |\log x|)\},$$

which proves (10). Divide (8) by the relevant variance expansion and expand the inverse of $\alpha \mapsto M_\nu(\alpha R)$. This gives (12)–(14). For $\nu > 1$, the Matérn differential equation and the recurrence

$$K_\nu(z) = K_{\nu-2}(z) + 2(\nu-1)K_{\nu-1}(z)/z$$

reduce the normalized-correlation coefficient to $A_{\nu,k}(r)$, proving its sign. Full transition constants and an exact exponential benchmark are in the reproducible technical manuscript.

At the remaining integer transition,

$$M_2(x) = 1 - \frac{x^2}{4} + x^4 \left\{ \frac{3}{64} - \frac{\log(x/2) + \gamma_E}{16} \right\} + O\{x^6(1 + |\log x|)\}.$$

For fixed $\nu > 2$, four-times differentiability yields $M_\nu(x) = 1 - x^2/\{4(\nu-1)\} + O(x^4)$. These formulas, and the fractional term for $1 < \nu < 2$, give the three stated smooth-regime remainders after division and inverse expansion.

4 FINITE-GRID SUPPORT-AWARE LIKELIHOOD

The theorem covers translation-invariant interior averaging. For a finite simulation, let X_{in} denote raw sites and let $S_h \in \mathbb{R}^{m \times n}$

be a rectangular row-stochastic observation operator. With noise present before smoothing,

$$y = S_h \{Y(X_{\text{in}}) + \epsilon\}, \quad \epsilon \sim N(0, \tau^2 I_n), \quad (19)$$

the support-aware covariance is

$$\Sigma_{\text{corr}}(\alpha) = S_h \{K_\alpha(X_{\text{in}}, X_{\text{in}}) + \tau^2 I_n\} S_h^\top. \quad (20)$$

The naive declared-center point-support covariance is instead

$$\Sigma_{\text{naive}}(\alpha) = K_\alpha(X_{\text{out}}, X_{\text{out}}) + \tau^2 I_m. \quad (21)$$

Besides ignoring support, (21) treats pre-smoothing noise as independent post-smoothing noise.

With variance, smoothness, and nugget fixed, the finite-design KL target minimizes

$$Q_n(\alpha) = \frac{1}{2} [\log \det \Sigma(\alpha) + \text{tr}\{\Sigma(\alpha)^{-1} \Sigma_0\}]. \quad (22)$$

The corrected family contains Σ_0 when the operator and fixed nuisance parameters are correctly specified and decay is identifiable; the naive family generally does not. Our primary experiment sets $\tau = 0$, isolating field support from noise-timing misspecification. A separate $\tau^2 = 0.05$ run is retained only as a combined-misspecification artifact. Rectangular S_h is intentional: an invertible square, parameter-independent transform may merely change coordinates, whereas aggregation reduces raw sites to fewer supported outputs.

Boundary row normalization is outside the interior theorem. If two location-dependent scaled kernels have means μ_s and μ_t , then

$$C_{h,s,t}(r) = C(r) + h \nabla C(r)^\top (\mu_s - \mu_t) + O(h^2).$$

A truncated kernel can therefore create a first-order term and destroy stationarity; the interior theorem gives no boundary sign guarantee. We include one boundary and one fixed irregular design as illustrations, not as a boundary scaling study.

5 SYNTHETIC VALIDATION

5.1 Continuous phase oracle

We used a two-dimensional product Epanechnikov density, $\alpha = R = 1$, tensor Gauss–Legendre order 96, 18 bandwidths from 0.003 to 0.3, and $\nu \in \{0.25, 0.5, 0.75, 1, 1.5, 2.5\}$. The density of each coordinate of D is available in closed form, so (3) is evaluated by deterministic quadrature. We solved (5) by monotone root finding. Repeating at orders 64 and 128 changed pseudo-decay by at most 4.4×10^{-8} and 5.2×10^{-9} , respectively, relative to order 96.

All 108 shifts $\alpha - \alpha_h^\dagger$ were positive. For smoothness 0.25, 0.5, and 0.75, log–log slopes over the six smallest bandwidths were 0.515, 0.999, and 1.468, versus theoretical powers 0.5, 1, and 1.5. The raw slope at $\nu = 1$ was 1.817, reflecting the logarithmic modifier. Slopes for $\nu = 1.5$ and 2.5 were 1.994 and 2.000. At the smallest bandwidth, ratios of the exact shift to the full theorem leading term were 1.021, 1.000, 0.958, 1.059, 0.997, and 1.000 in increasing order of ν .

5.2 Finite-grid full likelihood

The core design uses a 19×19 input lattice with spacing 0.25 and every second site as a candidate output center, trimmed by 0.75 in each coordinate. We set $\nu = \alpha = 1$, $\tau^2 = 0$, $\nu \in \{0.5, 1, 1.5, 2.5\}$,

Table 1: Finite-design results at $h = 0.7$. MCSE is for the reported mean.

ν	Model	Target	Mean	MCSE	Bound hits
0.5	Corrected	1.000	1.090	0.047	3
0.5	Naive	0.152	0.151	0.004	0
1	Corrected	1.000	0.990	0.010	0
1	Naive	0.414	0.407	0.005	0
1.5	Corrected	1.000	0.990	0.007	0
1.5	Naive	0.573	0.570	0.005	0
2.5	Corrected	1.000	0.994	0.004	0
2.5	Naive	0.740	0.738	0.003	0

and $h \in \{0, 0.3, 0.5, 0.7\}$. Radial Epanechnikov rows are normalized exactly. Stress designs retain boundary centers, perturb each coordinate uniformly in $[-0.04, 0.04]$ before clipping, and grow one-dimensional domains from 100 to 400 raw sites. The irregular panel is one fixed realization.

For each configuration, (22) was minimized by bounded scalar optimization over $\log \alpha \in [\log(0.02), \log(5)]$. Two hundred replicate likelihoods were evaluated on a 161-point log-decay profile; interior minima were refined by three-point quadratic interpolation and endpoint minima were retained as boundary fits. Gaussian raw-observation vectors were transformed by S_h . Common random numbers were reused across bandwidths. No adaptive diagonal jitter was added; a Cholesky or population-optimization failure would abort the configuration and make the reducer mark the run incomplete.

The immutable run contained 21 configurations and 8,400 fits. The reducer verified configuration hashes, row counts, and unique model–replicate keys; independent checks covered finiteness, model labels, replicate indices, and metadata. Across the core designs, corrected population targets were within 1.3×10^{-7} of one. At $h = 0.7$, naive targets were 0.152, 0.414, 0.573, and 0.740 as ν increased; Monte Carlo means were 0.151, 0.407, 0.570, and 0.738. Corrected means were 1.090, 0.990, 0.990, and 0.994. The first was within two Monte Carlo standard errors of its target. All six core boundary-hitting fits were retained.

Including boundary outputs changed both output dimension and the rough-field naive target, from 0.152 to 0.135 at $h = 0.7$; for the fixed jittered design the target moved from 0.289 to 0.284 at $h = 0.5$. Corrected targets remained one. In the domain-growth check, increasing raw sites from 100 to 400 reduced corrected-estimator standard deviation from 0.529 to 0.177 and naive-estimator standard deviation from 0.151 to 0.068, while at $n = 400$ their means were within 0.026 and 0.009 of their respective targets.

6 DISCUSSION

The theorem separates a standard operation from a less-studied inferential consequence. Propagating a covariance through a known support is classical; the new object is the parameter selected when that propagation is ignored. Rough Matérn fields move at a fractional bandwidth power because the origin cusp changes the variance more rapidly than fixed-lag covariance. Smooth fields move quadratically, but a Bessel identity still fixes the sign.

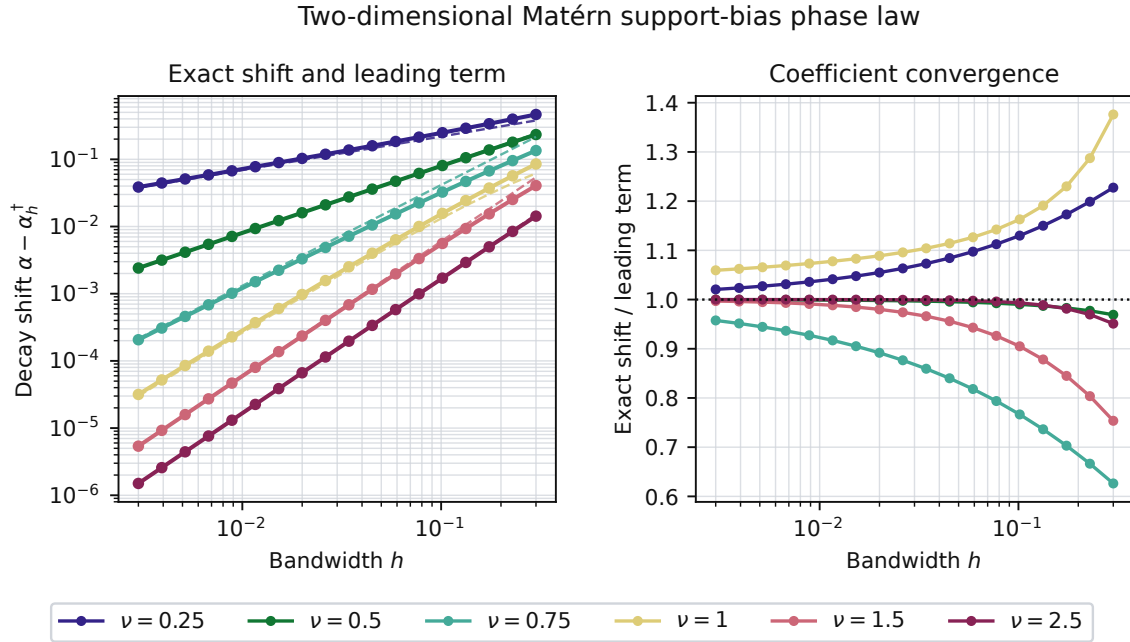


Figure 1: Continuous two-dimensional pseudo-decay shift. Left: exact shifts (solid) and theorem leading terms with explicit coefficients (dashed). Right: their ratio, which approaches one as $h \downarrow 0$.

The simulation has a deliberate role: it validates a theorem-linked generator, computes nonrandom finite-design KL targets, and distinguishes misspecification from finite-sample error. It is not a substitute for the proof. The zero-support case verifies exact agreement, the corrected target verifies model containment, and the boundary and irregular panels expose where the interior theorem stops.

The scope is narrow. Smoothness and support are known; nuisance parameters are fixed in the finite study. The closed target is for a two-point criterion, while full-grid targets are numerical. Fixed-resolution discrete smoothing does not converge to continuous convolution merely because the domain grows. Boundary normalization can introduce first-order terms. Joint support, smoothness, variance, and range estimation may be weakly identified. These are future problems, not implicit claims.

Recent finite-grid Matérn work studies discretization and edge effects without deriving this preprocessing-support pseudo-target phase law [9]. The defensible novelty claim is therefore limited to the explicit all- ν expansion and universal small-support range direction.

7 CONCLUSION

For a fixed nonzero-lag pair fit under translation-invariant interior averaging, ignored support makes a Matérn field look more persistent than it is. The inverse-range displacement is $h^{2\nu}$, $h^2 \log(1/h)$, or h^2 , and its leading sign is negative across fixed smoothness values, dimensions, and compact symmetric kernels. A support-aware covariance removes the finite-design population shift when the

known operator and fixed nuisance parameters are correctly specified and decay is identifiable. The result supplies a theorem-level diagnostic and a reproducible simulation protocol; it does not assert a universal sign for boundary supports or arbitrary full-likelihood fits.

Code, manifests, and audited aggregate results are available at <https://github.com/Sasha-Cui/HighDimSpatialStatistics> under the immutable tag `paper-draft-v0.1.0`.

REFERENCES

- [1] Takeshi Amemiya and Roland Y. Wu. 1972. The Effect of Aggregation on Prediction in the Autoregressive Model. *J. Amer. Statist. Assoc.* 67, 339 (1972), 628–632. <https://doi.org/10.1080/01621459.1972.10481264>
- [2] François Bachoc. 2018. Asymptotic Analysis of Covariance Parameter Estimation for Gaussian Processes in the Misspecified Case. *Bernoulli* 24, 2 (2018), 1531–1575. <https://doi.org/10.3150/16-BEJ906>
- [3] Erick A. Chacón-Montalván, Peter M. Atkinson, Christopher Nemeth, Benjamin M. Taylor, and Paula Moraga. 2024. Spatial Latent Gaussian Modelling with Change of Support. <https://doi.org/10.48550/arXiv.2403.08514> [stat.ME]
- [4] Maria Myrto Folia and Magnus Rattray. 2018. Trajectory Inference and Parameter Estimation in Stochastic Models with Temporally Aggregated Data. *Statistics and Computing* 28 (2018), 1053–1072. <https://doi.org/10.1007/s11222-017-9779-x>
- [5] Montserrat Fuentes. 2007. Approximate Likelihood for Large Irregularly Spaced Spatial Data. *J. Amer. Statist. Assoc.* 102, 477 (2007), 321–331. <https://doi.org/10.1198/016214506000000852>
- [6] Alan E. Gelfand, Li Zhu, and Bradley P. Carlin. 2001. On the Change of Support Problem for Spatio-Temporal Data. *Biostatistics* 2, 1 (2001), 31–45. <https://doi.org/10.1093/biostatistics/2.1.31>
- [7] Carol A. Gotway and Linda J. Young. 2002. Combining Incompatible Spatial Data. *J. Amer. Statist. Assoc.* 97, 458 (2002), 632–648. <https://doi.org/10.1198/016214502760047140>
- [8] Phaedon C. Kyriakidis and Eun-Hye Yoo. 2005. Geostatistical Prediction and Simulation of Point Values from Areal Data. *Geographical Analysis* 37, 2 (2005), 124–151. <https://doi.org/10.1111/j.1538-4632.2005.00633.x>

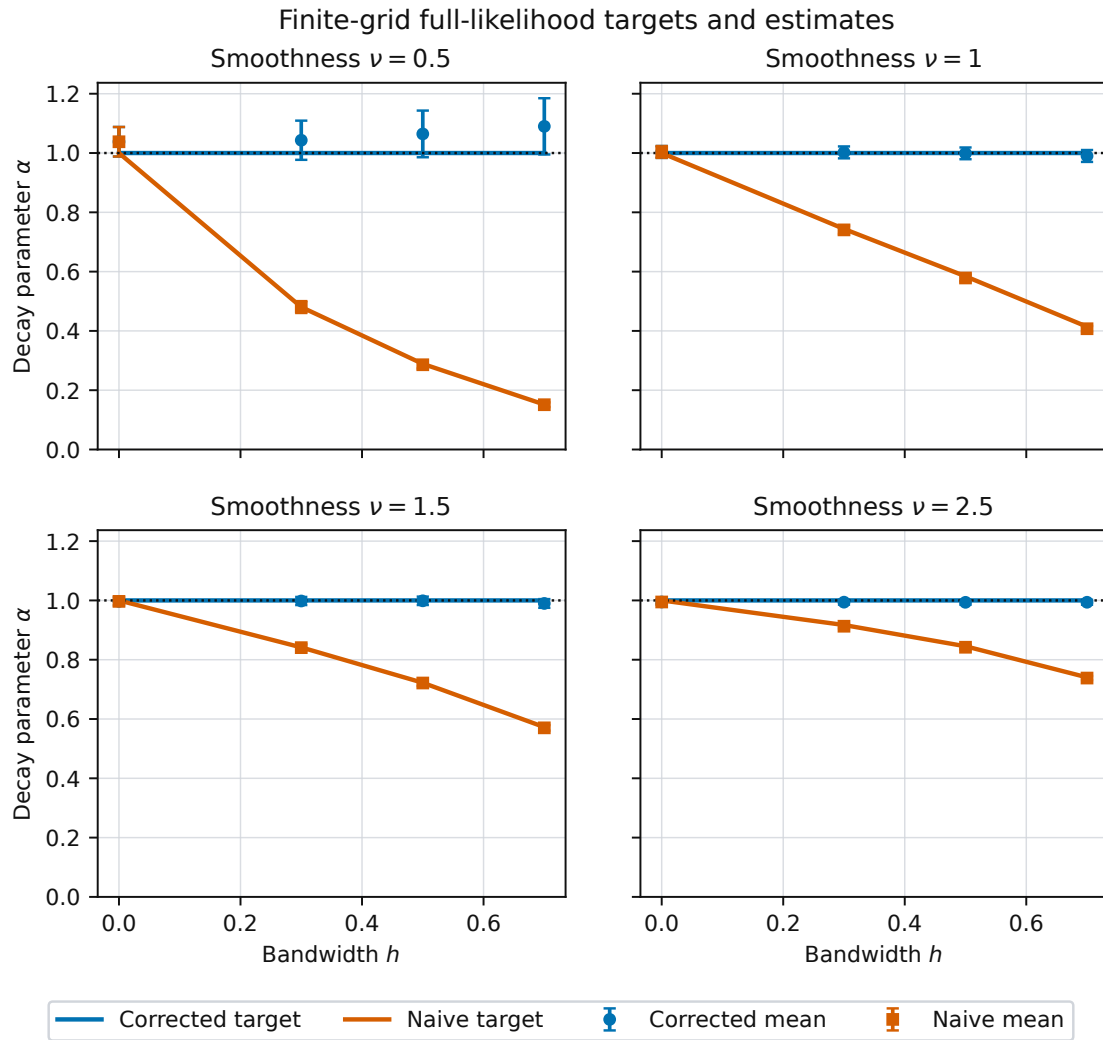


Figure 2: Finite-design full-likelihood targets and Monte Carlo mean estimates. Error bars show two Monte Carlo standard errors of the mean.

- [9] Frederik J. Simons, Olivia L. Walbert, Arthur P. Guillaumin, Gabriel L. Eggers, Kevin W. Lewis, and Sofia C. Olhede. 2026. Maximum-Likelihood Estimation of the Matérn Covariance Structure of Isotropic Spatial Random Fields on Finite, Sampled Grids. *Geophysical Journal International* 245, 2 (2026), ggag044. <https://doi.org/10.1093/gji/ggag044>

- [10] Daniel O. Stram and William W. S. Wei. 1986. Temporal Aggregation in the ARIMA Process. *Journal of Time Series Analysis* 7, 4 (1986), 279–292. <https://doi.org/10.1111/j.1467-9892.1986.tb00495.x>
- [11] Hao Zhang. 2004. Inconsistent Estimation and Asymptotically Equal Interpolations in Model-Based Geostatistics. *J. Amer. Statist. Assoc.* 99, 465 (2004), 250–261. <https://doi.org/10.1198/016214504000000241>